A Self-Calibrating Single-Camera Pressure Controller for Heterogeneous Urban Traffic: Bounded Fairness, Emergency Preemption and Microsimulation Evaluation

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***Abstract*—Fixed-time signal plans remain the default in many South Asian cities, where traffic is motorcycle-dominated, demand is strongly asymmetric between approaches, and a detector plant is unaffordable. We present RELAY, an adaptive signal controller whose only input is one existing fixed junction camera. A single convolutional detector yields per-approach, per-class counts, converted to passenger-car-unit (PCU) demand and consumed by a queue-weighted pressure controller in the max-pressure family with the operational rules a deployable signal needs, including a hard anti-starvation bound and emergency preemption with a bounded hold. The controller estimates each approach's arrival and discharge rate from the counts themselves, with no survey and no configured saturation flow, and uses them to size each green from Webster's cycle on the measured degree of saturation and to price a phase change at the discharge its clearance forfeits. It is evaluated against equal-split and demand-optimal Webster fixed plans, plain max-pressure, self-organising traffic lights and vehicle-actuated control in a seeded point-queue simulation, and against Webster, max-pressure and gap-actuated control in SUMO with a motorcycle-dominated fleet. In a point-queue grid of 72 cells spanning nine demand shapes on two topologies, mean delay falls against the Webster plan in every cell, by 1.4% in the weakest and by up to 43.8% in the single-junction experiment, and against actuated control in 64 of 72 cells. In SUMO the controller reduces delay in every one of 90 paired runs against each baseline, by 22.9% to 55.5% against Webster, although at the highest load its worst observed wait is longer than under either the Webster plan or actuated control. The controller is unaffected by the biased undercount the detector actually exhibits, the corridor coupling term contributes at most 3.9% of delay in either traffic model, and emergency preemption reaches green in 4.07 s on average over 900 trials with a worst case of 13.99 s against a closed-form bound of 14.0 s. This is a simulation and real-footage study, not a field deployment.**

***Index Terms*—Adaptive traffic signal control, max-pressure control, computer vision, heterogeneous traffic, emergency vehicle preemption, microsimulation, low-cost intelligent transport systems.**

I. Introduction

URBAN signal control in low- and middle-income cities is usually performed by fixed-time plans or by police officers operating signals by hand, neither of which measures what is queued: a busy arterial approach holds a long queue while the crossing minor street receives the green it would receive at three in the morning. Webster's method [1] showed that cycle length and splits should follow demand, but a fixed plan can only follow the demand measured when it was designed. Studies of signalised intersections in the Kathmandu valley report saturation flows and delays far from standard capacity guidance, attributed to the vehicle mixture and lane discipline [2], [3], exactly the condition under which an imported plan performs badly.

The established remedy is a demand-responsive system such as SCOOT [4] or SCATS [5], or an optimisation-based controller in the OPAC [6] and RHODES [7] lineage. They have decades of field evidence, but they assume a detector plant, communications and a control room, so per-junction cost is dominated by sensing and civil works; in a city where the signal head itself is sometimes unpowered, that capital profile is prohibitive.

Two conditions have changed. Fixed cameras are already installed at many urban junctions for enforcement and monitoring, so a camera-based controller reuses a paid-for asset, and single-stage object detectors now run at video rate on commodity hardware [8], [9], which makes per-approach counting a software problem. Vision-based queue estimation from fixed cameras is established [10], and several recent studies connect a detector to a signal timer on this basis [11]–[14].

What is often missing in that applied literature is the control side: many studies set green in proportion to a count, rarely state what prevents a minor approach from being starved, what happens when the detector fails or how the policy behaves at saturation, and compare only against a fixed-time plan, which any competent adaptive controller will beat. Conversely, the theoretical literature on max-pressure control [15] proves throughput optimality under ideal queue observation with no minimum green, no clearance interval and no fairness bound, all of which a deployable signal requires.

This paper describes and evaluates RELAY, a controller placed between those positions. Its contributions are:

- A single-camera perception-to-control pipeline for heterogeneous traffic in which presence and PCU-weighted demand are kept as separate quantities, because a PCU-weighted score alone classifies a lone motorcycle as an empty lane.
- A self-calibrating pressure controller. Arrival and discharge rates are estimated per approach from the count sequence alone (Section IV-C) and used for two decisions a constant cannot make: each stage's green is floored at

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its share of Webster's cycle on the measured degree of saturation (Section IV-D), and a stage change must beat the discharge its clearance interval forfeits (Section IV-E). No saturation-flow table, survey or per-site constant is required.
- Fairness enforced by explicit bounds rather than by the score: separate maximum-wait bounds for vehicles and pedestrians, a cap on the preemption hold, and an anti-starvation rule that serves the worst-treated queue instead of rotating among equally starved ones.
- An evaluation that reports where the controller loses: against a Webster plan computed from the true mean demand and three adaptive baselines (plain max-pressure, self-organising traffic lights and vehicle-actuated control), per cell over a 72-cell grid, and in SUMO with a motorcycle-dominated fleet, turning movements and finite links; the count error the detector actually makes, a biased density-dependent undercount, is modelled rather than replaced by zero-mean noise.

There is no field installation, no controller cabinet in the loop and no before-and-after study on a real signal: control results come from seeded simulation and perception results from recorded footage. The contribution is a design and feasibility study at the level a city could use to decide whether a pilot is worth funding; Section VII states which claims a field trial would still have to establish.

II. Related Work

A. Fixed-time, actuated and demand-responsive control

Webster's analysis of cycle length and green splits [1] remains the reference method for fixed-time plans and is the basis of the stronger fixed baseline used here. Vehicle-actuated control, in which a green is extended by a passage time on each detected arrival until a gap or a maximum is reached, is the most widely deployed adaptive scheme [16] and is our strongest adaptive baseline. SCOOT [4] adjusts splits, offsets and cycle time from flow profiles measured upstream of the stop line, SCATS [5] selects among plans using measured degree of saturation, and OPAC [6] and RHODES [7] solve a rolling-horizon optimisation over predicted arrivals. All depend on detectors installed and maintained in the carriageway, and all were designed for streams with lane discipline and similar vehicle sizes.

B. Max-pressure and self-organising control

Max-pressure signal control derives from backpressure scheduling in queueing networks [17]. Varaiya [15] showed that repeatedly activating the phase maximising a weighted difference between upstream and downstream queues stabilises any demand within the capacity region using only local queue information. Two properties of the theoretical policy conflict with practice: the argmax may switch as often as the decision interval allows, and each switch costs a clearance interval during which no approach discharges, which Levin et al. [18] address with a cyclical phase structure and our saturated-demand results quantify independently; and throughput optimality is proved for the unconstrained policy, so minimum green, maximum green, hysteresis and forced service void the proof. We therefore describe our controller as a queue-weighted pressure heuristic in the max-pressure family and make no stability claim. Self-organising traffic lights [19], [20] are a second decentralised rule that needs only local counts: a red phase accumulates waiting vehicles times seconds and takes green when the integral passes a threshold, unless the current green is discharging a short platoon. It is included as a baseline because it shares our information requirements and is simpler than our rule.

C. Learning-based control

Reinforcement learning is the dominant research direction for signal control; surveys by Wei et al. [21], Haydari and Yilmaz [22] and Noaeen et al. [23] cover the field, and PressLight [24] derives its reward from max-pressure. We did not adopt a learned policy, for reasons specific to the deployment context: training requires a simulator calibrated to the target junction and none exists for the junctions we are concerned with; a neural policy does not naturally expose the maximum-wait or clearance guarantees an authority approving a signal plan needs; and it transfers poorly across geometries, whereas a pressure rule parameterised only by a phase set moves to a new junction by drawing polygons on a camera image. The comparison we do not make, and which should be made, is a head-to-head against a trained policy on the same arrival streams.

D. Camera-based counting and vision-driven signal control

Vehicle detection from fixed cameras is a mature application of single-stage detectors [8], [9]. Our system uses the Ultralytics implementation of the YOLO family [25]; the architecture generation we fine-tune is described in [26], which remains a preprint. Because the controller consumes instantaneous per-approach occupancy rather than cumulative crossings, we count detections inside approach polygons instead of tracking vehicles, which removes identity-switch errors in dense queues from the control path.

Recent applied studies pair such a detector with a signal timer. Papanashi et al. [11] report YOLO-based adaptive timing evaluated in Bengaluru, a heterogeneous-traffic context comparable to ours; Alruyshid et al. [12] add emergency and bus prioritisation in a Baghdad case study; Talaat et al. [13] and Medvei et al. [14] report detector-driven controllers, the latter combining detection with a learned policy. These works establish that the perception component is practical; our contribution relative to them is on the control and evaluation side: an explicit pressure formulation, stated fairness bounds, a preemption latency bound, and an evaluation against demand-optimal fixed-time and actuated baselines that reports where the adaptive policy loses.

Emergency preemption has a long operational history and a careful literature on its side effects, since a preemption interrupts the cycle and can create hazards at the transition [27]; Humagain et al. [28] review route optimisation and preemption methods. Our detection channel differs: the emergency vehicle

is recognised visually by the same detector that counts ordinary traffic and no vehicle-side equipment is required, cheaper and less reliable, which is why the ambulance class carries the highest confidence threshold and the hold is capped; what we add is a worst-case latency bound with a measurement against it.

III. System Architecture

RELAY consists of a perception stage and a control stage connected by one data structure, a mapping from approach identifier to per-class vehicle count; everything downstream is independent of how the counts were obtained, so the same controller runs on recorded footage, a live camera, a synthetic feed or a microsimulator.

A. Perception

A single fixed camera views the junction. Each approach is a polygon in normalised image coordinates, drawn once per site; no per-site model training is performed. For each frame a YOLO detector produces class-labelled boxes, filtered in four steps. First, per-class confidence gating, because two-wheelers are small in the image and are lost at a threshold that suppresses false positives on cars: 0.24 for motorcycles and bicycles, 0.38 by default, 0.50 for trucks and 0.68 for ambulances. The high ambulance gate is a safety choice, since an ambulance detection moves the signal by itself, and a vehicle below the gate still contributes ordinary demand through its next-best label. Second, cross-class deduplication: a greedy pass in descending confidence retains a box only if its intersection over union with every retained box is below 0.5, since per-class non-maximum suppression can leave one distant vehicle carrying stacked car, truck and motorcycle boxes. Third, approach assignment by centroid containment, followed by an exponential moving average per approach and class with coefficient 0.35, roughly a one-second time constant at the observed detection rate. Fourth, failure handling: a bad frame returns the last smoothed counts, and a staleness flag set after three seconds without an update feeds the controller empty counts, so the wait clocks and the maximum-wait rule cycle the phases safely instead of holding green on stale demand.

B. Control, topology and deployment

The controller consumes the count mapping and a set of conflict-free phases, and is topology-agnostic: four-arm, three-arm and two-arm junctions differ only in the phase set passed at construction, and approaches may be specified per arm or per movement, in which case the phase set is generated to include paired through phases, paired protected-turn phases and one split phase per arm. Pedestrian demand enters as a separate input giving, per arm, the number waiting, the longest wait and the number crossing; it is sanitised before use (unknown arms, negative counts and non-finite waits are discarded, counts clamped to 200 and waits to 180 s) so a client fault cannot inject an unbounded aging term. The intended installation is one small computer per junction, taking a feed from an existing camera and driving the existing signal controller, with no central server in the control path. We measure the perception cost in Section VI-E but claim no cost figure, since no hardware was procured.

IV. Control Formulation

A. Notation and derived quantities

Let A be the set of approaches, P a set of phases (each $p \in P$ a subset of A served without conflict) and Δt the control step. Each step perception reports counts $n_{a,c}$ for approach a and class c, and the approaches on which an emergency vehicle was detected. Two quantities are derived, the PCU-weighted demand and the occupancy indicator:

$$d_a = \sum_c w_c \, n_{a,c}, \qquad o_a = \mathbb{1}\left[\textstyle\sum_c n_{a,c} > 0\right], \tag{1}$$

with weights w_c of 1.0 for a car, 0.3 for a motorcycle, 2.5 for a bus or truck, 2.0 for an ambulance, 0.2 for a bicycle and 0.8 for an auto-rickshaw: configuration values for a motorcycle-dominated mix, ordered consistently with published equivalency estimates for South Asian traffic [29], [30] rather than estimated locally, with their sensitivity measured in Section VI-I. Keeping these separate matters: a single motorcycle contributes $d_a = 0.3$, which a demand threshold would treat as an empty approach, so the empty-phase skip and gap-out rules key on o_a and a lone two-wheeler still receives a full minimum green.

Each approach carries a clock r_a answering how long the oldest waiting vehicle has waited, not how long since the approach last had green. With G the set of approaches currently green, z_a the consecutive seconds the approach has read empty and $\tau_e = 3$ s,

$$r_a \leftarrow \begin{cases} r_a\,(1-\delta_a) & a \in G, \\ r_a + \Delta t & a \notin G,\ o_a = 1, \\ 0 & a \notin G,\ o_a = 0,\ z_a \geq \tau_e, \\ r_a & \text{otherwise}, \end{cases} \tag{2}$$

where δ_a is the fraction of the approach's demand that discharged during the step. On green the clock unwinds by the fraction of the queue that discharged and reaches zero only when the queue does; resetting it on the green would claim that everything still behind the stop line had stopped waiting, which beyond capacity is false. The third case stops a long-vacant approach accumulating a clock that the first arriving vehicle inherits, and the debounce keeps a detection dropout from erasing a real queue's guarantee.

B. Phase score

Define a bounded urgency multiplier $u_a = 1 + \min(r_a, R)/R \in [1, 2]$, with R the maximum-wait parameter. The vehicle pressure of a phase is

$$D(p) = \sum_{a \in p} d_a \, u_a. \tag{3}$$

Aging multiplies each approach's own demand and is capped at a factor of two, so waiting cannot manufacture demand that

does not exist; an additive term $d_a + \lambda r_a$ inverts the intended ordering (with $\lambda = 0.4$ a two-vehicle approach that has waited 45 s scores like an eighteen-PCU queue), and a phase-level $\max_a r_a$ lets a starving single-vehicle turn bay double the score of a phase whose large through queue has just been served.

Pedestrians are treated as demand. A phase p frees the crossing of arm x when neither x nor the opposing arm, whose straight-ahead movement traverses the same crossing, is served: $F(p) = \{x : x \notin \text{arms}(p),\ \text{opp}(x) \notin \text{arms}(p)\}$. With m_x people waiting to cross arm x, k_x currently on the crossing and v_x the longest pedestrian wait,

$$\Phi(p) = w_{\text{ped}}\, M(p)\Big(1 + \frac{\min(\bar{v}(p),\ V)}{V}\Big), \tag{4}$$

where $M(p) = \sum_{x \in F(p)} (m_x + k_x)$ counts the people the phase would serve, $\bar{v}(p)$ is the longest wait among them, $w_{\text{ped}} = 0.3$, so one waiting person contributes about as much pressure as one motorcycle, and V the pedestrian maximum-wait parameter. The complete score adds a large emergency constant $B(p) = B_0 \mathbb{1}[\exists a \in p : a \in \hat{E}]$ with $B_0 = 10^6$:

$$S(p) = D(p) + \Phi(p) + B(p). \tag{5}$$

Detectors drop frames, so each approach carries a hold timer reset to $\theta = 2.5$ s whenever an emergency is announced on that arm and otherwise decaying by Δt; the latched set is $\hat{E}$, and every movement of an arm is latched together because the camera cannot resolve the lane.

C. Online estimation of arrival and discharge

Both remaining rules need to know how fast an approach fills and how fast it empties, which a fixed-time plan takes from a survey and a saturation-flow table. We estimate both from the count sequence the detector already produces, which keeps the controller free of per-site constants.

Let $d_a(t)$ be the PCU demand on approach a at step t. An approach on red cannot discharge, so every increase in its queue is arrivals; an approach on green moves by arrivals less departures, so departures are the arrival rate less the observed change:

$$\hat{\lambda}_a \leftarrow \hat{\lambda}_a + \alpha\big(\Delta_a(t) - \hat{\lambda}_a\big), \tag{6}$$

$$\hat{s}_a \leftarrow \hat{s}_a + \alpha\big(\hat{\lambda}_a - \Delta_a(t) - \hat{s}_a\big), \tag{7}$$

where $\Delta_a(t) = (d_a(t) - d_a(t-1))/\Delta t$, (6) is applied on a step during which the approach was red, (7) on a step during which it was green with $d_a(t-1) > 0$, $\alpha = \Delta t/\tau$, and both estimates are clamped below at zero. Three details matter. The change between two counts was produced under the signal state of the step that has just ended, so the sample is attributed to that state and not to the one being decided; attributing it to the current state charges the first green step with a red step's growth and measured 0.49 PCU/s against a true 0.55. The clamp is applied to the running estimate and never to the sample, because clamping samples discards only the low side of the noise and inflates $\hat{s}_a$ to 0.79. And (7) is conditioned on the queue held before the step, not after it, since conditioning on the latter deflates the estimate to 0.30. With all three the estimator recovers 0.51 to 0.60 PCU/s against a truth of 0.55, and $\hat{\lambda}_a$ to within roughly 20%; both start at zero, so the two rules below are inert for the first few cycles.

D. Green duration from the measured cycle

Webster [1] minimises delay at cycle length $C = (1.5L + 5)/(1-Y)$, where L is the lost time per cycle and $Y = \sum_p y_p$ sums the critical flow ratios $y_p = \max_{a \in p} \lambda_a / s_a$, allocating each phase a green of $(C - L)y_p/Y$. Equations (6) and (7) supply y_p directly, so the allocation normally applied offline to survey data is evaluated continuously from the junction in front of the camera:

$$g^\star(p) = \max\Big(g_{\min},\ (C - L)\frac{y_p}{Y}\Big), \tag{8}$$

$$C = \min\Big(C_{\max},\ \frac{1.5L + 5}{1 - Y}\Big), \tag{9}$$

with $C = C_{\max}$ when $Y \geq 1$. Y must sum the critical ratios of one cycle, so phases are accumulated heaviest-first over disjoint approach sets, since on a movement-level junction the phase sets overlap. Webster's C diverges as $Y \to 1$ and does not exist beyond it; the clamp at $C_{\max} = 120$ s, the longest cycle a fixed plan would be given, bounds the floor and removes the discontinuity. An earlier version fell back to $g_{\max}$ there, and with a noisy Y near 1 the floor swung between greens of over 200 s and 45 s; Section VI-F reports the cell where this was found.

$g^\star(p)$ is evaluated once, when the phase takes green, and acts as a floor on that green in place of $g_{\min}$ in rule 4 of the stage machine; evaluating it continuously would chase a target that falls as the queue drains. It is a floor and not a hold: preemption and the empty-phase skip are evaluated above it and still use $g_{\min}$. Under light demand Y is small and the rule never binds. It is deliberately not capped at $g_{\max}$, which is a fairness bound on how long one movement may hold the junction; refusing a green the measured demand requires does not protect the minor approaches, because the uncleared queue is still there at the next cycle.

E. Pricing a phase change

A phase change costs $y + \rho$ seconds during which no approach discharges. The approaches currently green would have spent that interval discharging, so the challenger must beat the incumbent by more than what the handover forfeits:

$$\kappa(p) = \sum_{a \in p} \min\big(d_a,\ \hat{s}_a(y + \rho)\big), \tag{10}$$

and rule 9 of the stage machine switches only when $S(p^\star) > S(p) + h + \kappa(p)$. The minimum caps the forfeit at what the approach still has queued, so a drained or empty phase costs nothing to leave; a constant margin h cannot express this, being the same whether the junction is empty or saturated.

F. Target selection and the stage machine

The signal cycles through green, yellow of duration y and all-red of duration ρ; a green is never followed directly by a conflicting green. The next phase is chosen by strict

precedence. *(i) Emergency:* if $\hat{E} \neq \emptyset$, take the phase serving the most emergency approaches, breaking ties by the largest r_a among them. *(ii) Vehicle starvation:* otherwise, if any approach has $r_a \geq R$ and $d_a > 0$, force a phase serving the worst such approach, preferring one that also serves approaches within 25 s of the threshold so that two clearance intervals are not paid where one would do. Candidates are restricted to approaches not currently green whose r_a exceeds that of every approach the current phase serves: the rule serves the worst-treated queue and does not rotate, since beyond capacity every approach sits past R permanently. *(iii) Pedestrian starvation:* otherwise, if any arm has $m_x > 0$ and $v_x \geq V$, force a phase in $F(x)$ with the greatest pedestrian demand. *(iv) Score:* otherwise take $\arg\max_p S(p)$ over phases with demand, resting in all-red if none has.

A running green is terminated by the first applicable rule, where e is elapsed green, γ_t the gap-out time, Θ the preemption cap, W the walk-hold allowance and h the hysteresis margin:

1) An emergency exists on another phase and not the current one, and either $e \geq g_{\min}$ or the current phase is empty.
2) An emergency exists on the current phase: hold, except that an approach reaching $r_a \geq 2R$ forces service and the cumulative hold is capped at Θ, after which the emergency term is dropped.
3) The current phase is empty (no served approach has $o_a = 1$, no freed crossing has pedestrian demand) while another phase has demand: switch immediately.
4) $e < g^\star(p)$: hold, with $g^\star(p) \geq g_{\min}$ allocated at onset by (8).
5) Vehicle starvation elsewhere, then pedestrian starvation elsewhere: switch.
6) A pedestrian is still on a crossing freed by the current phase and $e < \min(g_{\max}, g_{\min} + W)$: hold; the bound stops a renewing pedestrian stream pinning green on a lightly used approach.
7) Gap-out: the served approaches have read empty for at least γ_t and another phase has demand.
8) Maximum green: $e \geq g_{\max}$: yield to the best other phase with demand.
9) Score against the cost of switching: $S(p^\star) > S(p) + h + \kappa(p)$ for the best alternative $p^\star$.

Parameter values are listed in supplementary Table S1.

G. Corridor coupling and the preemption bound

On a corridor each junction j knows, for some arms, the downstream approach (j', a') that the arm's discharge feeds. The demand presented to the score is discounted by the downstream queue $q_{j',a'}$,

$$\tilde{d}_a = \begin{cases} \max\left(d_a - \gamma\, q_{j',a'},\ \varepsilon\right) & o_a = 1, \\ 0 & o_a = 0, \end{cases} \tag{11}$$

with $\gamma = 0.4$ and $\varepsilon = 0.05$, where the floor at ε keeps the coupling a discount on pressure rather than an erasure of presence. This is a one-sided approximation of the movement-level pressure difference of max-pressure theory, using the aggregate downstream queue because it is the only downstream quantity one camera per junction can supply.

The worst case from first emergency detection to green is an announcement that arrives just after the controller has committed to a transition away from the phase that serves the emergency: the transition completes, the new phase must be given its minimum green, and only then can the preemption take effect through a second clearance interval:

$$L_{\max} = y + \rho + g_{\min} + y + \rho. \tag{12}$$

With the single-junction parameters (supplementary Table S1), $L_{\max} = 14.0$ s. The system also ships an executable self-check asserting twelve properties (clearance, green allocation, fairness, emergency handling and input validation), all passing on the version evaluated; these are assertions under specified stimuli, not proofs.

V. Experimental Setup

A. Point-queue simulation and baselines

Most control results are obtained in a point-queue simulation, in which delay accumulates at a single point rather than along a link. While an approach displays green it discharges at $s = 0.55$ veh/s, a saturation headway of about 1.8 s within the range capacity guidance gives for a through lane [31], accumulated as a fractional carry so a green shorter than one headway releases no vehicle, with no banking of capacity while the queue is empty. Arrivals are Poisson per approach, drawn once per step from a single seeded generator and pushed into every policy's queues, so the comparison is paired at the level of individual vehicles. Runs are 1800 s with the first 300 s discarded, and vehicles still queued at the end are counted with the wait accumulated so far. The controller under test is the system's own code on identical arrival streams; the cost is traffic-model fidelity, which Experiments 8 and 9 address in SUMO [32].

Five baselines are used. The *equal-split* plan is a round-robin giving each phase 13.0 s of green, a 35 s cycle with 3.0 s yellow and 1.5 s all-red. The *Webster* plan takes cycle and splits from Webster's formulae computed from the *true* mean demand of the run: with lost time $L = |P|(y + \rho)$ and critical flow ratios summing to Y, the cycle is $(1.5L + 5)/(1 - Y)$ clamped to $[30, 120]$ s, effective greens $(C - L)y_i/Y$ floored at $g_{\min}$, and when $Y \geq 0.95$ the cycle is set to the upper clamp with proportional splits; it is deliberately generous to fixed-time control, being granted perfect knowledge of the average demand and re-derived for every cell. *Max-pressure* is Varaiya's policy as usually simulated: at every step after the minimum green, take the phase with the largest queue, with a clearance interval and no other rule. *SOTL* is the self-organising rule of [19], [20] with threshold $\theta = 40$ vehicle-seconds and platoon cut-off $\mu = 3$. *Actuated* is gap-based vehicle-actuated control [16]: green is extended by a 3 s passage time on each arrival up to the maximum green and ends on gap-out, and phases are served in sequence, skipping phases without a call. All adaptive baselines use the same $g_{\min}$, $g_{\max}$, yellow and all-red as the controller.

B. Microsimulation

Experiments 8 and 9 use SUMO 1.27.1 [32] through its TraCI interface. The single junction has four 300 m arms of two lanes each; vehicles go straight with probability 0.8 and turn left or right with probability 0.1 each, across the opposing stream under permissive right of way. The fleet is 62% motorcycles, 30% cars, 5% buses and 3% trucks by count, with widths narrowed so two-wheelers share a lane and the sublane model [33] at 0.8 m lateral resolution so they filter through gaps; departures are Poisson per arm at the point-queue rates. The controller sees only what a camera would, vehicles by class within 80 m of each stop line read every 0.5 s, never SUMO's internal queue, and its decision is imposed on the SUMO traffic light each step. The Webster baseline is computed from the realised PCU demand of each run and a saturation flow of 1531 PCU/h/lane measured in the same simulation behind a standing queue; the actuated baseline is SUMO's own gap-actuated program with a 3 s maximum gap and the same green bounds. Delay is SUMO's per-vehicle time loss against free flow over vehicles departing after a 300 s warm-up in 1800 s runs, 10 seeds per cell.

C. Experiments

Experiment 1: single junction. A four-arm junction with two phases under three demand profiles: *asymmetric*, $(0.150, 0.130, 0.040, 0.035)$ veh/s for N, S, E and W; *balanced*, 0.089 veh/s on all four; *single*, 0.250 veh/s on one approach and 0.035 veh/s on the others; each scaled by $m \in \{0.6, 0.8, 1.0, 1.2, 1.4\}$, giving 767 to 1794 veh/h, 30 seeds per cell, with paired t and Wilcoxon signed-rank tests on the per-seed reductions. Experiment 1b measures phase change rate and displayed-green fraction over 10 seeds per cell.

Experiment 2: three-junction corridor, point queues. A straight arterial carrying a three-arm, a four-arm and a second three-arm junction at 90 m spacing, with the topology, phase sets, routes and timing of the system's network demonstration; link travel time 7.2 s at 12.5 m/s. Four policies: the controller with coupling (11), the same with $\gamma = 0$, an independent Webster plan per junction, and a coordinated Webster plan with a common cycle and eastbound progression offsets. Runs are 2400 s with 400 s warm-up, 20 seeds, three demand levels, and one emergency vehicle released from the west at $t = 1200$ s.

Experiment 3: preemption latency. The junction is driven to a loaded state for a random 40 to 160 s, then an emergency vehicle is announced on a random approach and re-announced every step; the delay until that approach displays green is recorded and the identical trial replayed against the equal-split plan, which has no preemption path; 300 trials at each of three demand levels.

Experiment 4: detector. The shipped detector on the system's held-out split. **Experiment 5: footage.** The full stack over 60 s of two real fixed-camera clips at 960 px on an Apple M3 Pro, and the count as a function of inference resolution with smoothing disabled.

Experiment 6: robustness grid. Seven demand shapes on the four-way junction plus two on a three-arm T, each at eight load levels running past the point where a fixed plan saturates: 72 cells of 25 paired seeds, reported per cell; repeated with 20% zero-mean Gaussian miscounting (6a) and under 10 biased count models matching the detector's actual failure (6b): a saturating undercount $q/(1 + q/q_{1/2})$ with $q_{1/2} \in \{8, 4\}$, flat losses of 20, 40 and 60%, and the same losses with a queue of one vehicle rounding to zero, with and without a floor of one.

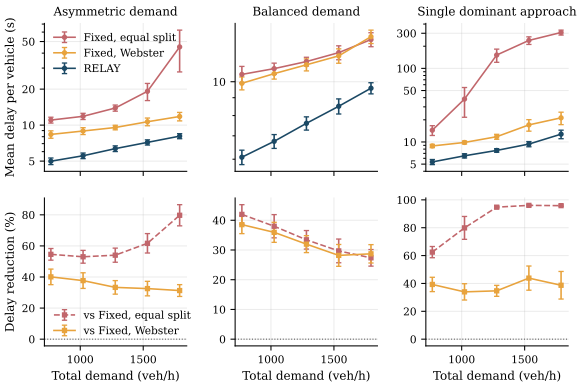


Fig. 1. Experiment 1. Top: mean delay per vehicle against total demand for the three profiles, logarithmic axis. Bottom: paired per-seed reduction against each fixed-time baseline; error bars one standard deviation over 30 seeds. Supplementary Table S2 lists every cell.

Experiment 7: adaptive baselines. The grid with max-pressure, SOTL and actuated control on the same arrivals.

Experiment 8: SUMO single junction. The three Experiment 1 profiles at $m \in \{0.6, 1.0, 1.4\}$, 10 seeds, against Webster, max-pressure and SUMO's actuated control.

Experiment 9: SUMO corridor. The Experiment 2 arterial rebuilt in SUMO with 150 m links, so a downstream queue can physically block an upstream discharge; the four policies of Experiment 2, 10 seeds, three levels.

Experiment 10: sensitivity. The fairness bound R from 30 s to unbounded on the single-dominant shape at the two highest loads, and the motorcycle PCU weight from 0.2 to 1.0 with a class-mixed queue whose two-wheeler share differs by axis (80% on N–S, 30% on E–W).

VI. Results

A. Single junction

Fig. 1 gives the single-junction results, with per-cell values in supplementary Table S2. RELAY reduces mean delay against both baselines in all 15 cells, by 28.2% to 43.8% against the Webster plan and by 27.4% to 96.0% against the equal-split plan, every difference significant at $p < 0.05$ by the Wilcoxon signed-rank test (largest $p = 0$). Under asymmetric demand, the regime the system is designed for, the reduction is 54.0% against equal split and 33.3% against Webster at 1278 veh/h. The gain against equal split grows with demand, reaching 79.7% at 1789 veh/h, but that growth is a capacity artefact: there the equal-split plan is oversaturated on the major approaches and its delay diverges, whereas the Webster plan and RELAY remain stable at essentially the same throughput, 1804 and 1801 veh/h. The gain against Webster is close to flat across the range, 40.1% down to 31.3%, and

we regard the Webster column as the honest headline. Under a single dominant approach the equal-split plan collapses (at 1278 veh/h a mean delay of 151.8 s and a throughput of 1123 veh/h against the demand offered); against Webster the gain peaks at 43.8% at 1534 veh/h and is 38.7% at the highest load. Under balanced demand, the hard case for a pressure policy, the gain against Webster is 38.5% at 769 veh/h and 28.7% at 1794 veh/h.

The wait distribution (supplementary Fig. S1) is where the fairness bound becomes visible. At 1278 veh/h under asymmetric demand the 95th-percentile wait is 29.9 s under equal split, 25.8 s under Webster and 14.5 s under RELAY, with worst observed waits of 50.0, 50.1 and 26.7 s. Under a single dominant approach the contrast is larger: 333.4 and 36.3 s at the 95th percentile against 17.0 s, and worst waits of 361.5 and 72.9 s against 36.6 s, consistent with the forced-service rule at $R = 60$ s. The tail improves in every cell, which matters more than the mean for a policy whose fairness claim is a bound.

B. Switching, clearance and what saturation costs

Balanced saturated demand is where an unconstrained pressure policy fails, and the failure motivates three of the rules in Section IV. No phase dominates, successive scores sit near-tied, a constant hysteresis margin is crossed on noise, and an anti-starvation rule that fires whenever any approach has waited past R fires on every step, because beyond capacity every approach is past R permanently. Both terminate green at $g_{\min}$: measured on this junction at 2307 veh/h, a policy with those two properties displayed 5.2 s of green per 4.5 s of clearance, held green for 0.54 of wall time against the fixed plan's 0.80, and reached 94 s of mean delay against that plan's 19 s, the practical form of the issue Levin et al. [18] address with a cyclical structure. Sections IV-D and IV-E recover the lost capacity with a cycle measured rather than configured: under balanced demand RELAY still switches more often than the fixed plan (Experiment 1b, supplementary Fig. S2), 283 to 331 per hour against 204, and displays green for less of the wall clock, 0.585 to 0.644 against 0.743, while reducing delay by 28.2% to 38.5%, because a fixed plan spends its green where the demand is not. Under a single dominant approach at $m = 1.4$ RELAY switches less than the fixed plan, 167 per hour, holding green for 0.790 of the time.

C. Corridor, point queues

Experiment 2 (supplementary Table S3) gives three findings, one negative. The delay reduction decays only gradually with demand: 57.9%, 49.4% and 44.8% against independent Webster plans at 769, 1282 and 1794 veh/h, and 56.2%, 47.8% and 42.3% against the coordinated plan, with through travel time and stops improving together with delay (32.3 s and 1.87 stops against the coordinated plan's 39.3 s and 2.03 at the highest demand). The coupling term does essentially nothing, changing mean delay by +0.1%, -0.4% and +0.2% when disabled, because a point queue has no spatial extent and a downstream queue never physically blocks an upstream discharge; Experiment 9 asks the same question with links that can fill. The emergency vehicle traverses the corridor in 22.3 to 26.7 s under RELAY against a free-flow 21.6 s and 30.1 to 53.6 s under independent Webster plans; the coordinated plan manages 24.0 to 31.5 s because the vehicle travels in the progression direction, whereas RELAY is direction-agnostic.

D. Emergency preemption latency

Mean latency is 4.07 s against 7.00 s for the fixed cycle over 900 trials (supplementary Table S4), almost insensitive to demand (4.02 s at the lightest load, 4.16 s at the heaviest) because preemption is evaluated above the measured-cycle floor against $g_{\min}$, so a busier junction does not make an emergency vehicle wait longer. The tail is the important quantity: the 95th percentile is 11.33 s against 20.59 s and the worst case 13.99 s against 22.29 s, just below the bound of 14.0 s from (12); every trial above 13.4 s is one in which the announcement arrived within the first second of a yellow already transitioning away from the emergency's own phase. Because the bound is composed only of clearance and minimum-green settings, it can be stated in advance to an emergency service. Two qualifications: in 34.6% of trials the emergency approach already had green when the vehicle was detected, and in 33.1% the fixed cycle happened to reach that approach sooner than the preemption did, so the distributions overlap (supplementary Fig. S3) even though the tails differ sharply. The claim these data support is a bounded worst case, not a uniformly faster response.

E. Perception

On the held-out split of 97 images and 4990 labelled instances, the 2.59-million-parameter detector reaches a mean average precision of 0.954 at IoU 0.5 and 0.809 averaged over IoU 0.5 to 0.95, with precision 0.950 and recall 0.886; per-class AP at IoU 0.5 (supplementary Fig. S4) exceeds 0.96 for cars, buses, trucks and ambulances and is lowest for motorcycles at 0.851, the smallest and most occluded class. The split mixes automatically labelled frames from a synthetic feed with real frames pseudo-labelled by a general detector, so a high score establishes that the fine-tuned model reproduces the labelling function it was trained against, not accuracy against human ground truth.

On real footage at 960 px, median per-frame latency is 46.8 ms on the boulevard clip (21.4 frames per second, 95th percentile 85.7 ms) and 87.6 ms on the dense mixed-traffic clip (11.4 frames per second), well above what a control loop with a 5 s minimum green needs.

The main perception weakness is visible on the dense clip (supplementary Fig. S5): a few dozen vehicles are detected in a scene containing several hundred. Against inference resolution (supplementary Fig. S4) the mean sampled count on that clip rises from 21.4 at 640 px to 26.5 at 960 px, 33.2 at 1280 px and 39.0 at 1600 px, an increase of 83% with no sign of convergence (motorcycles alone 120%), on the sparser boulevard clip from 4.85 to 10.00, at a median inference cost rising from 47.2 to 108.0 ms. A count that keeps growing with resolution has not converged, so the absolute demand the controller receives on dense two-wheeler scenes is an

underestimate of unknown magnitude, worse where the queue is denser. This does not immediately invalidate the control results, because the controller compares approaches rather than reading absolute volumes; Experiment 6b measures how much it matters.

F. Robustness across demand shapes, topology and count error

Table I reports the wider grid. RELAY reduces mean delay against both fixed-time baselines in 72 of 72 cells spanning 432 to 2462 veh/h, and never falls below 26.9% against the equal-split plan. The worst cell against Webster is the single-dominant shape at 2300 veh/h, where the reduction is 1.4%: 61.2 s of mean delay against 61.4 s, at throughputs of 2221 and 2238 veh/h. There the dominant approach alone demands 82% of all displayed green, the demand-optimal cycle is over 260 s, and neither controller serves the demand.

This cell is worth dwelling on because an earlier version of the controller lost it by 26.6%, and the explanation we first gave, the fairness bound R, was wrong: Experiment 10 swept R from 30 s to unbounded and the delay did not move (Section VI-I). The mechanism was in (8): with the measured Y fluctuating around 1, the earlier fallback to $g_{\max}$ above $Y = 1$ alternated the floor between a green of over 200 s and one of 45 s, the controller ran a shorter effective cycle than the 120 s plan, and at a degree of saturation of 0.93 the extra 3% of time in clearance was the entire margin. Clamping the measured cycle at $C_{\max}$, as the fixed plan is clamped, brought the cell to within a few per cent of the plan with no other cell worse; we report the sequence because it is the kind of error a per-cell evaluation exists to catch.

Repeating the grid with 20% zero-mean Gaussian miscounting leaves the picture almost unchanged, 71 of 72 cells won with a worst of -8.1% against Webster, the one loss the beyond-capacity cell again; that test is weak, because the estimators smooth over a $\tau = 60$ s window that averages zero-mean noise away. Experiment 6b applies the error the detector actually makes (supplementary Table S5). Under a saturating undercount the controller wins 71 and 71 of 72 cells with mean reductions of +36.0% and +35.3%, the single loss in each being the beyond-capacity cell, where compressing a long queue distorts the saturation estimate. A flat loss of 20 or 40% changes nothing (71 and 72 cells), because the degree of saturation in (8) is a ratio of estimates drawn from the same counts and the phase comparison in (5) is invariant to a common scale. A loss of 60% collapses to 51 cells, but not because of the scale: at that factor a queue of one vehicle rounds to zero, and the models in which a lone vehicle reads as none fail the same way (50 cells) whatever their scale, because the empty-phase skip and the wait clock are keyed on presence. A presence floor of one restores every one of them (72, 71 and 72 cells). The requirement this yields for the perception stack is narrow and testable: an unbiased presence signal at short queues, not accurate absolute counts.

G. Adaptive baselines

Experiment 7 (supplementary Fig. S6) adds the adaptive baselines on the same grid. Against plain max-pressure RELAY wins 72 of 72 cells with a worst of $+9.1\%$ and a median of 20.8%; against SOTL 72 cells, worst $+5.8\%$, median 12.1%. Vehicle-actuated control is the serious competitor: RELAY wins 64 of 72 cells with a median reduction of 9.3% and loses 8, all at or beyond capacity on balanced or near-balanced shapes, by at most 1.3%. The mean advantage is modest; the difference lies in the tails. Averaged over the grid, the 95th-percentile wait is 20.2 s under RELAY against 24.1 s under actuated control, 49.6 s under max-pressure and 61.4 s under SOTL, and the worst wait 35.4 s against 36.2, 74.9 and 77.1 s. Max-pressure and SOTL have no fairness bound and their tails show it; actuated control's sequential service gives it one, so the honest summary is that RELAY is an actuated controller that needs no detectors in the road, sizes its greens from a measured cycle, and carries preemption.

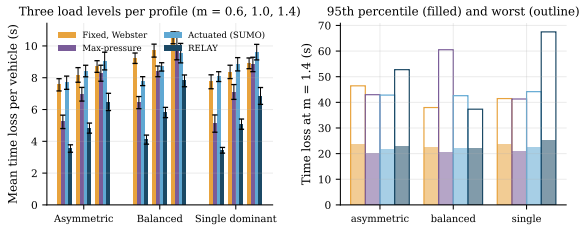


Fig. 2. Experiment 8. Left: mean time loss per vehicle in SUMO by policy and demand. Right: 95th-percentile and worst time loss at the highest load, where RELAY's tail is no longer the shortest.

H. Microsimulation

Fig. 2 gives the microsimulation result, with per-cell values in supplementary Table S6. RELAY reduces mean time loss in every one of the 90 paired runs against each of the three baselines: by 22.9% to 55.5% against the Webster plan, 21.5% to 36.1% against max-pressure, and 18.1% to 57.4% against SUMO's actuated program. The margin over actuated control is larger than in the point-queue harness, and the reason is visible in the traffic: with a motorcycle-dominated fleet a gap-actuated detector sees a continuous stream of small headways and extends green to its maximum on the busy approach, whereas the measured-cycle floor sizes the green to the degree of saturation and hands over sooner. The reduction falls with load in every profile, as it does in the point-queue grid.

One result goes the other way. At the highest load the worst observed time loss is 52.5 s under RELAY against 42.0 s under Webster and 43.2 s under actuated control, averaged over the three profiles, and the 95th percentile is 23.3 s against 23.2 s. In the point-queue harness the fairness bound held the tail; in SUMO it does not, because the bound applies to what the camera sees. A vehicle beyond the 80 m detection zone is on no wait clock, and under saturation the queue extends past the zone, so the vehicles that wait longest are precisely the ones the controller cannot count. This is a real limitation of camera-based control with a fixed field of view, and the strongest argument here for a wider or second camera at a junction expected to saturate. At light and moderate load, where the queue sits inside the zone, the tail is shortest under RELAY (11.8 s against 22.4 s at the lightest load).

TABLE I
EXPERIMENT 6. WORST AND MEDIAN PAIRED DELAY REDUCTION OVER THE EIGHT LOAD LEVELS OF EACH DEMAND SHAPE, 25 SEEDS PER CELL. THE WORST COLUMN IS THE REPORTABLE ONE.

Topology	Shape	Worst vs equal	Worst vs Webster	Median vs Webster
Four-way	Asymmetric	+53.2	+29.0	+33.7
	Balanced	+26.9	+26.9	+30.5
	Near-equal	+27.9	+27.7	+30.8
	One axis	+63.4	+29.2	+37.3
	Three heavy	+32.9	+34.1	+35.0
	Lopsided	+52.6	+33.8	+35.4
	Single	+61.3	+1.4	+37.7
T	Balanced	+36.8	+36.2	+38.8
	Arterial	+63.1	+30.4	+38.4

The SUMO corridor (Experiment 9, supplementary Table S7) settles the coupling question the point-queue corridor left open. With links short enough that a downstream queue physically blocks an upstream discharge, disabling the coupling term changes mean time loss by +1.5%, +1.4% and -3.9% at the three levels, at most 3.9%. The corridor gain over the coordinated Webster plan, 31.2% to 67.6%, is therefore attributable to local adaptivity alone in both traffic models; the aggregate downstream queue is too weak a proxy for the movement-specific pressure difference to add anything measurable, and the design would lose nothing by dropping it.

I. Sensitivity

Experiment 10 (supplementary Fig. S7) gives the two sensitivities. The fairness bound R does not move the mean at all on the single-dominant shape: from 30 s to unbounded, mean delay stays between 12.7 and 12.7 s at 1789 veh/h and between 61.1 and 61.7 s at 2300 veh/h, with the reduction against Webster between +0.9% and +1.5% at the higher load, and the worst wait is likewise flat, 80 to 80 s, because at this shape the measured-cycle floor, not the starvation rule, determines when the minor approach is served; R binds where the score would otherwise leave a small queue unserved, the light-load case of Section VI-A. The motorcycle weight, with two-wheeler shares that differ by axis, changes the reduction against Webster by at most 0.4 percentage points across the three shapes (asymmetric +32.6% to +32.9%); the controller degrades gracefully because it compares phases rather than reading absolute volumes and the measured-cycle floor is a ratio in which a per-class weight partly cancels.

VII. LIMITATIONS

This is not a field study. No signal was controlled, there is no cabinet in the loop and no before-and-after measurement, so the failure modes that appear only in the field (power interruption, lens contamination, camera repositioning, seasonal light, drivers who learn that the signal responds to them) are untested.

The fairness bound applies to the visible queue. Experiment 8 shows that at saturation the vehicles waiting longest are beyond the detection zone. Any camera-based controller shares this, and it should be stated to an authority as a condition on camera placement rather than hidden in a mean.

Perception was not validated against human ground truth. The held-out split mixes synthetic frames with pseudo-labelled real frames, so mAP measures agreement with the labelling process; Experiment 6b bounds the consequence for control, but the magnitude of the undercount on the target city's junctions is unknown and a human-annotated set is required before any absolute demand claim.

Narrower gaps. The microsimulation uses SUMO defaults and a fleet plausible for the region rather than one fitted to a Kathmandu junction, so it establishes that the advantage survives car-following, lane changing, turning conflicts and spillback, not its magnitude at any site. The evaluated junctions have two stages, so movement-level phasing and protected turns are generated but not exercised; a learned policy such as PressLight [24] is not compared; the pedestrian logic is asserted by executable invariants but never compared against a conventional pedestrian stage; and the corridor coupling term should be removed or replaced by a movement-level pressure difference.

VIII. CONCLUSION

We have described a single-camera adaptive signal controller for heterogeneous urban traffic and evaluated it in a seeded point-queue simulation, in SUMO with a motorcycle-dominated fleet, and on real fixed-camera footage. Its distinguishing feature is that it measures the junction rather than being told about it: arrival and discharge rates estimated from the same counts the detector produces size each green from Webster's cycle on the measured degree of saturation and set the margin a competing phase must clear.

Against a fixed plan granted the true mean demand the controller reduces mean delay in every cell of the primary grid and in every one of the 72 cells of the robustness grid, the weakest of them near parity beyond capacity; it beats plain max-pressure and self-organising control everywhere, and vehicle-actuated control in 64 of 72 cells with a markedly shorter tail. In SUMO it reduces time loss in every paired run against Webster, max-pressure and actuated control, by 22.9% to 55.5% against the Webster plan, while its worst-case wait at saturation is longer than under either that plan or actuated control, because the fairness bound reaches only as far as the camera sees. Two negative results are reported: the corridor

coupling term contributes nothing in either traffic model, and an explanation we first offered for the one weak cell was wrong and is replaced by a measured one. The next steps follow from the limitations: a human-annotated perception set from the target city, a comparison against a learned policy on identical arrivals, and a field trial.

Data, Code and Declarations

The controller, perception stack, simulation and SUMO harnesses, experiment scripts, raw per-run result files and the figure and table generators are available at https://github.com/kafle1/relay under the AGPL-3.0 licence; every number in this manuscript is a macro generated from the raw result files by `paper/experiments/make_numbers.py`, and `paper/experiments/verify_claims.py` regenerates and checks them. The author declares no competing interests and received no funding. No human subjects or personal data were involved: the footage is publicly available recorded video of public roads, used only for aggregate counting, with no identification of individuals or vehicles attempted.

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